

VIVEK VIJAYKUMAR INGLE

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SUMMARY

Robotics AI Engineer specializing in robot learning, visual imitation learning, computer vision, and robotic manipulation. Experienced in developing AI-driven robotics systems spanning perception, trajectory generation, simulation, model evaluation, and ROS 2-based robot execution. Current work focuses on World Action Models and their application to visual imitation learning, simulation-based robot learning, and industrial robotic manipulation.

TECHNICAL SKILLS

Programming: Python, C++, C#, MATLAB, JavaScript

Robot Learning: World Action Models, Robot Learning, Imitation Learning, Skill Reuse

AI/ML: PyTorch, TensorFlow, Deep Learning, Computer Vision, Model Evaluation, Edge AI

Computer Vision: OpenCV, YOLO, RGB-D Processing, Depth/Point Clouds, Object Detection, Object Pose Estimation, Camera-to-Robot Transformations, SLAM

Robotics: ROS/ROS 2, Robot Manipulation, Pick-and-Place, Robot Control, Cartesian Trajectories, Motion Planning, Sensor Fusion, Robot-Camera Calibration, Gripper Integration

Simulation & Evaluation: MuJoCo, RViz, Gazebo, Headless GPU Evaluation, Research Reproduction

Robotics Systems: Fast DDS, Vulcanexus, Docker, Linux/Ubuntu, Git, GitHub, CI/CD

Development: Streamlit, REST APIs, OPC UA

PROFESSIONAL EXPERIENCE

MYWAI, Sestri Levante, Italy

AI & Robotics Engineer

November 2025 – Present

- Design and implement advanced AI algorithms for industrial robotic automation, specializing in Imitation Learning techniques for pick-and-place applications with 95%+ task success rate
- Implement end-to-end robotics AI workflows spanning data collection, trajectory generation, model evaluation, and robotic execution
- Integrate computer vision, robot control, learned trajectories, and robotics software into manipulation pipelines
- Investigate modern robot-learning approaches, including World Action Models and visual imitation learning, and evaluate their applicability to robotic manipulation
- Develop and debug ROS 2-based robotics pipelines for perception, trajectory handling, robot control, and system integration
- Work with simulation, GPU-based evaluation, Docker-based environments, and Linux robotics infrastructure to support reproducible experimentation

MYWAI, Sestri Levante, Italy

Robotics Software Engineer (Internship)

May 2025 – November 2025

- Developed ROS 2-based robotic control software supporting imitation-learning and robotic manipulation workflows
- Designed and implemented full-stack Streamlit-based Imitation Learning toolkit enabling streamlined experimentation and data visualization for robotics researchers
- Successfully delivered MVP earning transition to full-time role; toolkit now used across organization for rapid prototyping and model evaluation

Heidelberg University, Heidelberg, Germany

Erasmus+ Research Trainee (Master's Thesis)

October 2022 – March 2023

- Designed and implemented haptic immersive VR scenarios with bimanual robotic exoskeleton for stroke rehabilitation under Prof. Lorenzo Masia at ARIES Lab
- Developed robotic mirror therapy system combining VR exergames with real-time haptic feedback, improving patient engagement scores by 35%
- Implemented control algorithms for multi-DOF robotic exoskeleton systems with emphasis on adaptive and safe human-robot interaction.

KEY PROJECTS

I-GENIUS – VISUAL IMITATION LEARNING & ROBOTIC MANIPULATION | *Python, ROS2, VILMA, Streamlit, YOLO, Docker, Fast DDS, Vulcanexus*

- Developed and integrated a visual imitation-learning pipeline for robotic pick-and-place, connecting demonstration data, trajectory generation, semantic action representations, skill reuse, and downstream robot execution
- Implemented a vision-based manipulation pipeline using ZED RGB/depth and point-cloud data with YOLO-based object detection and object pose estimation for perception-driven robotic tasks
- Implemented eye-to-hand camera calibration and camera-to-robot coordinate transformations, integrating learned Cartesian trajectories with ROS 2 for robot-relative manipulation and execution
- Built and debugged the end-to-end AI I-Genius pipeline from visual demonstration and perception through trajectory generation, ROS 2 communication, and execution on a Fairino FR10 industrial robot, using Fast DDS, Vulcanexus, and Docker for system integration

Monocular SLAM with TensorFlow Object Detection | *Python, C++, TensorFlow, ORB-SLAM2, OpenCV*

- Engineered portable 3D environment mapping system combining monocular SLAM (ORB-SLAM2) with real-time object detection using TensorFlow, utilizing only standard smartphone camera to eliminate need for specialized depth sensors or LiDAR
- Achieved real-time performance with <30ms processing time per frame, enabling low-cost spatial mapping and object recognition suitable for resource-constrained mobile robotics and AR applications
- Implemented robust feature tracking and loop closure detection algorithms, improving mapping accuracy by 25% in challenging lighting conditions and dynamic environments
- Validated system performance through comprehensive real-world testing scenarios including indoor navigation, object localization accuracy measurements, and multi-session map consistency evaluation

Human Activity Recognition for Activities of Daily Living (ADL) | *Python, Scikit-learn, SVM, Sensor Fusion, Linux*

- Developed a distributed Human Activity Recognition system using data from 6 IMU sensors to classify 9 Activities of Daily Living for assistive robotics
- Implemented an end-to-end machine learning pipeline for data preparation, normalization, SVM-based multiclass model training, testing, and per-sensor activity prediction
- Designed a sensor-fusion architecture using majority voting across six independently trained IMU classifiers to generate robust final activity predictions for assistive robotics and human-support applications

EDUCATION

Master's Degree in Robotics Engineering

September 2019 – July 2024

Università degli Studi di Genova, Genoa, Italy

Erasmus+ Exchange: Heidelberg University, Germany (October 2022 – March 2023)

Bachelor of Technology in Electronics & Media Technology

2012 – 2016

Karunya Institute of Technology and Sciences, India

Diploma in Game and Interactive Media Design

2017 – 2018

FX School

CERTIFICATIONS & ACHIEVEMENTS

- **Erasmus+ Scholar:** Selected for competitive robotics engineering exchange program at Heidelberg University
- **Google Cloud:** Innovating with Google Cloud AI Certification
- **Industry Experience:** Electronic Arts Software Engineering Virtual Experience Program
- **Additional Certifications:** C# & Unity Games Engine, Exploratory Data Analysis with MATLAB